UNIVERSITY ENGAGEMENT PROGRAM 5.0

HealthConnect

Final Project Submission

TEAM NAME HealthConnect Team	SUBMISSION DATE 31 August 2026
REPOSITORY github.com/kofdarc/pharma-link	LIVE WEBSITE healthconnect.dev

TEAM MEMBERS

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Student IDs: to be supplied before submission

Healthcare, finally connected.

1 Final Report

Project Overview

HealthConnect is a connected healthcare marketplace and operations platform for Lebanon. It allows patients to find medicines that are genuinely available, obtain and manage secure prescriptions, order from one or more pharmacies, and follow delivery progress. Doctors can issue QR-based electronic prescriptions. Pharmacies can manage stock, sales, clients, billing, imports, prescription fulfillment, and analytics. Drivers receive optimized pickup and delivery routes.

Problem Statement

Medicine discovery in Lebanon is fragmented. Patients often call multiple pharmacies without knowing which location has stock. Pharmacies operate isolated inventory systems and cannot see demand that never reaches the counter. Paper prescriptions make verification and partial dispensing difficult. Orders split across pharmacies also create duplicate trips, higher delivery cost, and slower service. These gaps affect patients, pharmacists, physicians, and delivery teams, particularly when time-sensitive medication is needed.

Solution Overview

HealthConnect connects the participants without requiring pharmacies to abandon their existing software. The platform ranks verified availability, plans the smallest practical pharmacy set for each basket, reserves stock, and consolidates pickups into efficient routes. Secure QR prescriptions support controlled access and partial dispensing. Operational dashboards and ledger-derived analytics help pharmacies act on stock, expiry, margin, and unmet-demand data.

Core value

FOR PATIENTS One search, transparent availability, secure prescriptions, and coordinated fulfillment.	**FOR PHARMACIES** A connected sales channel, interoperable inventory, and decision-ready analytics.
FOR DOCTORS Fast, verifiable prescribing with controlled dispensing and renewal workflows.	**FOR DELIVERY TEAMS** Consolidated, constraint-aware routes and a focused driver workflow.

Development Process

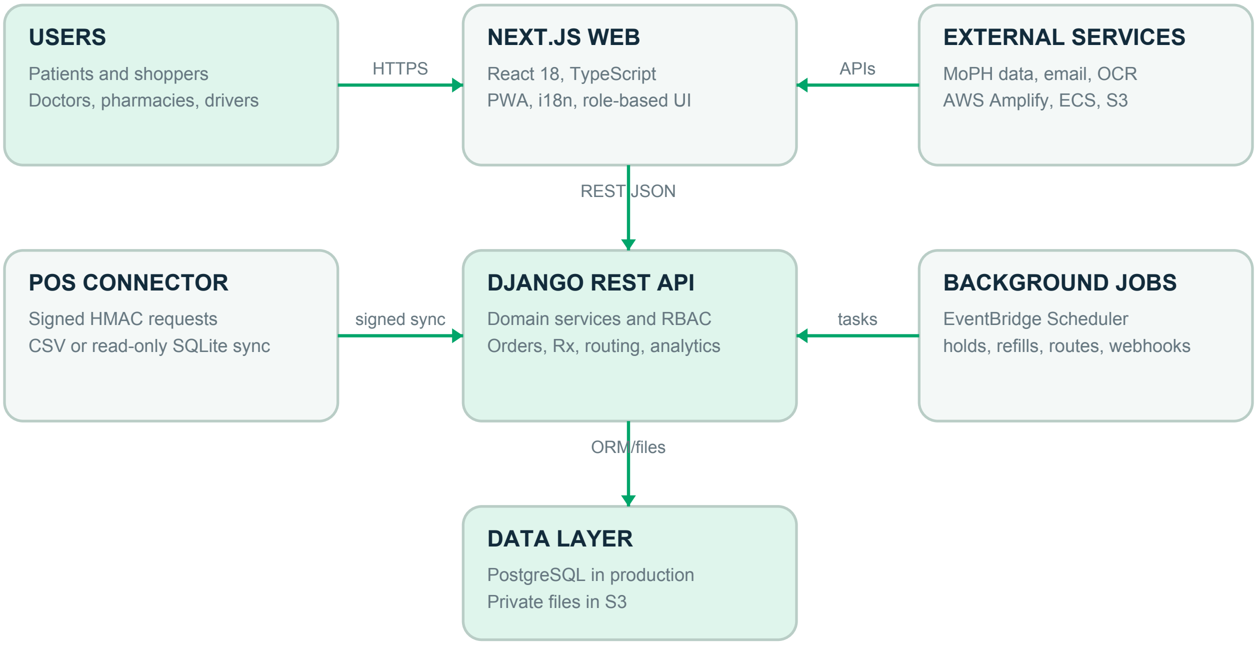
The Git history shows an iterative, milestone-based approach from initial requirements and monorepo scaffolding through domain implementation, integration, localization, deployment, and repeated production hardening. Work was integrated through feature commits, branches, and pull requests on the main branch.

1. Discovery and foundation	Product requirements, Django and Next.js monorepo, core domain boundaries.
2. Marketplace and operations	Medicine search, pharmacy inventory, sales, orders, delivery, and role-based workflows.
3. Clinical and integration flows	Electronic prescriptions, patient records, insurance, messaging, and the POS connector.
4. Intelligence and experience	Routing optimization, pharmacy analytics, OCR, localization, PWA support, and UI refinement.
5. Production readiness	AWS deployment, CI checks, scheduled operations, monitoring, security fixes, and build verification.

Technical Stack

Category	Technology / Tool
Frontend	Next.js 14 App Router, React 18, TypeScript 5.6, PWA, responsive CSS
Backend	Python 3.12+, Django 5, Django REST Framework 3.15, Gunicorn, WhiteNoise
Database	PostgreSQL in production, SQLite for local and isolated test workflows
Data and files	Amazon S3, openpyxl, xlrd, Pillow, QR generation, private prescription storage
Integration	Standard-library Python POS connector, HMAC-signed API, CSV and read-only SQLite inputs
Hosting	AWS Amplify for web, ECS Express Mode for API, ECR, RDS, EventBridge Scheduler, SQS DLQ
Delivery	Docker, Docker Compose, GitHub Actions, digest-pinned container releases
Version control	Git and GitHub
Other	OCR provider abstraction, MoPH catalog sync, email, analytics provider abstraction

Architectural Design Diagram



Features Implemented

01	Cross-pharmacy medicine discovery Searches brand, generic name, composition, and aliases. Results expose orderable availability without revealing exact pharmacy stock.
02	Basket sourcing and reservations Uses weighted set-cover logic to minimize unnecessary splits, then holds FEFO stock safely during fulfillment.
03	Secure electronic prescriptions Doctors issue QR prescriptions with hashed secrets, access logging, lockout controls, partial dispensing, and renewal support.
04	Prescription OCR intake Pharmacies and patients can upload paper prescriptions for structured extraction through configurable OCR providers.
05	Multi-pharmacy ordering Coordinates pharmacy acceptance, preparation, handover, invoicing, shopper tracking, recurring refills, and reviews.
06	Delivery route optimization Consolidates shared pharmacy pickups while enforcing precedence, capacity, opening times, and delivery windows.
07	Pharmacy operations Provides batch inventory, imports, POS synchronization, counter sales, invoices, clients, ledgers, billing, and staff workflows.
08	Analytics and decision support Calculates turnover, DIO, GMROI, ABC classes, dead stock, expiry exposure, reorder points, and unmet demand from source ledgers.
09	Role-based platform Supports administrators, pharmacy owners and staff, doctors, shoppers, and drivers in English, Arabic, and French, including RTL.
10	Production delivery Uses CI, AWS-hosted web and API services, scheduled background processing, monitored releases, and documented rollback procedures.

Challenges Faced and Solutions

Challenge 1: Reliable multi-pharmacy routing

DESCRIPTION	SOLUTION
A partially assembled route appears infeasible because a dropoff cannot occur until all pickups exist. Early feasibility checks therefore rejected valid multi-pharmacy jobs.	The solver now scores partial states by distance and checks constraints only after a complete candidate is assembled. Regret-ordered insertion and Or-opt relocation improve the plan, while tests pin precedence, capacity, time-window, consolidation, and baseline behavior.

Challenge 2: Preserving failure evidence across transactions

DESCRIPTION	SOLUTION
Large atomic transactions rolled back evidence that needed to survive exceptions, including prescription brute-force counters and unmet-demand records.	Transaction scopes were narrowed. Failure logs and counters are committed outside the block that raises, while stock and dispensing mutations remain atomic. Regression tests protect both behaviors.

Challenge 3: Integrating existing pharmacy systems safely

DESCRIPTION	SOLUTION
Pharmacies use different product codes and cannot pause operations for a full migration. Connector delays can also make old stock observations look current.	A standard-library connector maps local SKUs once, sends only changed absolute levels through signed and replay-resistant requests, preserves idempotency, flags reservation shortfalls, and penalizes stale POS observations during sourcing.

Team Contributions

The following summary is inferred from repository history and should be confirmed by the team before submission. Student IDs were not available in the repository.

John Kofdarelli	Project foundation and requirements; backend and product features; billing; patient and prescription workflows; OCR and analytics; delivery and search fixes; AWS infrastructure, CI/CD, monitoring, documentation, branding, and production hardening.
Marc Nadim Zeenny	Marketplace and pharmacy operations; role-based frontend workflows; POS connector; architecture and demo documentation; cross-platform compatibility; Amplify SSR build work; MoPH catalog and medicine imagery; medicine-detail experience; authentication fixes; UI and visual refinements.

2 Final Code Submission

GITHUB REPOSITORY	https://github.com/kofdarc/pharma-link
REVIEWED REVISION	d88a6c3cea77a468408199c6e106cd1612f80980
BRANCH	main

Repository Readiness

Check	Status	Evidence
Code organization	**PASS**	Clear monorepo structure: apps/api, apps/web, tools/connector, docs.
README and setup	**PASS**	README links architecture, API, setup, demo, AI, and AWS deployment guides.
Dependencies	**PASS**	Python and Node requirements are versioned and documented. Environment examples and Docker Compose are included.
Frontend type-check	**PASS**	pnpm exec tsc --noEmit completed successfully on 31 August 2026.
Frontend production build	**PASS WITH WARNINGS**	Next.js compiled, checked, and generated 70 static pages. Seven non-blocking lint warnings remain.
Backend tests	**NOT RE-RUN**	This checkout has no installed Python environment containing Django. The repository documents test commands, but this report does not present an unverified current pass count.
Committed and pushed	**CONFIRM BEFORE SUBMISSION**	The final remote synchronization state was not changed or asserted by this report.

Known Technical Limitations

- Travel time uses a distance approximation rather than live traffic routing.
- No real Lebanese digital payment provider is connected. Cash on delivery and a mock gateway exercise the interface.
- Route re-planning is scheduled or manually triggered. Drivers do not receive live WebSocket pushes.
- The POS connector has no installer or operating-system service wrapper.
- Additional database-level PII encryption and production Postgres volume encryption require infrastructure completion.

3 Working Demo

VIDEO DEMO LINK TO BE SUPPLIED. Upload to Google Drive and enable: Anyone with the link can view.
DEPLOYED WEBSITE https://healthconnect.dev

Recommended Two-Minute Demo Flow

0:00-0:15	Problem and roles Open the landing page. Explain that HealthConnect connects patients, pharmacies, doctors, and drivers across Lebanon.
0:15-0:40	Medicine search Search for a medicine. Show verified availability, regulated price labeling, pharmacy ranking, and the orderable quantity ceiling.
0:40-1:05	Basket and sourcing Add items from different pharmacies. Open the basket and show the sourcing plan and its explanation before placing the order.
1:05-1:30	Prescription Show a doctor-issued QR prescription, then open the public dispense flow and demonstrate controlled partial dispensing.
1:30-1:50	Routing Open the dispatch board. Highlight shared pickup stops, optimized distance, and avoided pharmacy visits.
1:50-2:00	Outcome Close on the pharmacy analytics dashboard and state the value: faster access, connected inventory, and fewer delivery trips.

Recording Checklist

- Keep the final recording at or below two minutes.
- Use seeded data and rehearse every login and transition before recording.
- Show complete user flows, not isolated static screens.
- Use a stable 1080p capture with readable browser zoom and no notification overlays.
- Remove login delays, loading pauses, setup terminals, and failed attempts.
- Verify the Google Drive sharing setting in a signed-out browser window.